

Git Hands-On: Connecting Local Repository to GitHub

Objective

Connect a local Git repository to a remote GitHub repository, push all local commits, and verify that the repository is successfully synchronized.

Step 1: Navigate to the Repository

```
cd ~/GitDemo  
pwd  
git status
```

Observed Output:

- Changed to the GitDemo directory.
- Displayed the current working directory.
- Verified the working tree was clean.

Step 2: Verify Local Branch

```
git branch -a
```

Observed Output:

- Displayed the local master branch before configuring the remote.

Step 3: Check Remote Repository

```
git remote -v
```

Observed Output:

- Initially no remote was configured.

Step 4: Add the GitHub Remote

```
git remote add origin https://github.com/RekhaS2005/GitDemo.git  
git remote -v
```

Observed Output:

- Added the origin remote.
- Verified fetch and push URLs.

Step 5: Push Repository to GitHub

```
git push -u origin master
```

Observed Output:

- Successfully uploaded all commits and configured master to track origin/master.

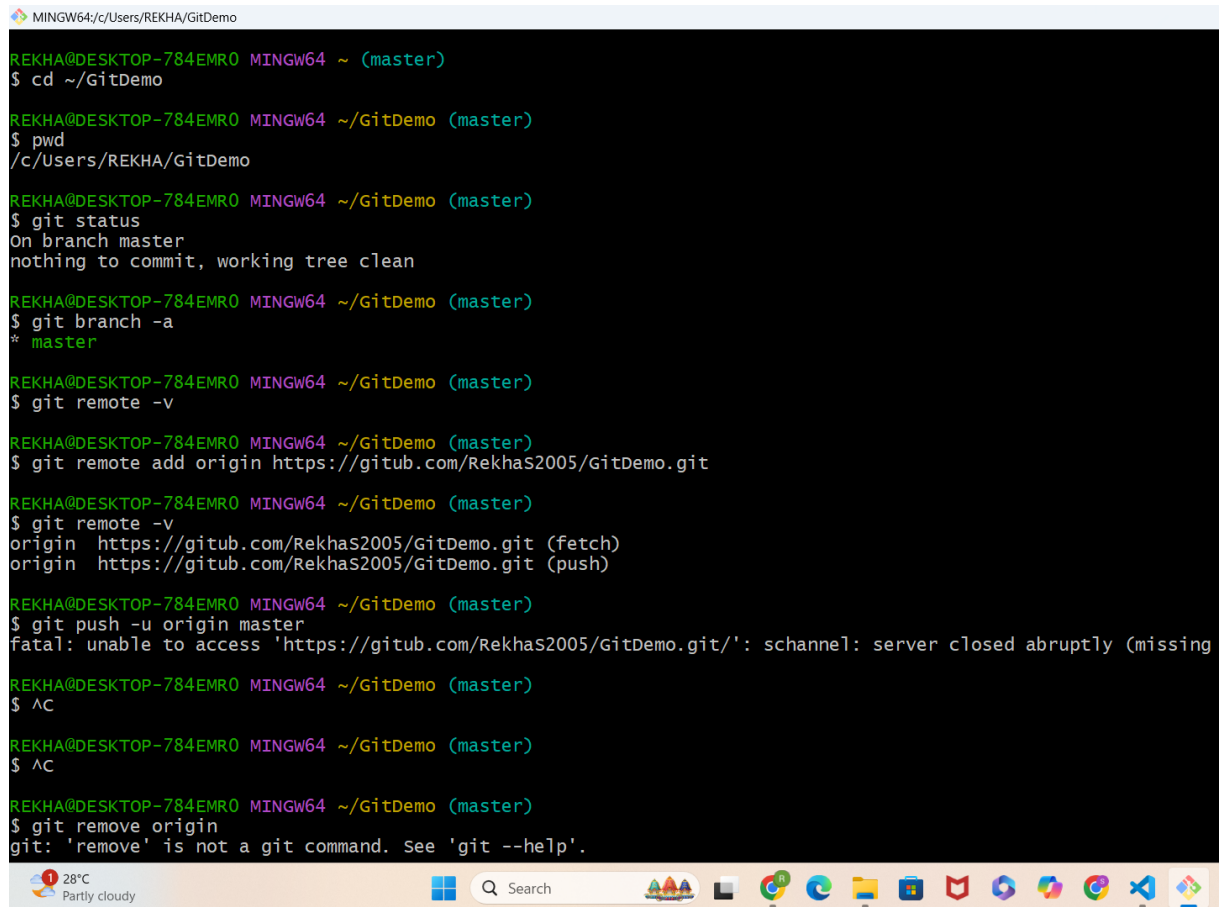
Step 6: Verify Synchronization

```
git branch -a
git log --oneline --graph --decorate --all
```

Observed Output:

- Verified remotes/origin/master exists.
- Verified commit history and origin/master reference.

Screenshots



The screenshot shows a Windows terminal window with the title bar 'MINGW64/c/Users/REKHA/GitDemo'. The terminal displays the following commands and output:

```
REKHA@DESKTOP-784EMR0 MINGW64 ~ (master)
$ cd ~/GitDemo

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ pwd
/c/Users/REKHA/GitDemo

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch -a
* master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git remote -v

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git remote add origin https://gitub.com/Rekhas2005/GitDemo.git

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git remote -v
origin https://gitub.com/Rekhas2005/GitDemo.git (fetch)
origin https://gitub.com/Rekhas2005/GitDemo.git (push)

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git push -u origin master
fatal: unable to access 'https://gitub.com/Rekhas2005/GitDemo.git/': schannel: server closed abruptly (missing

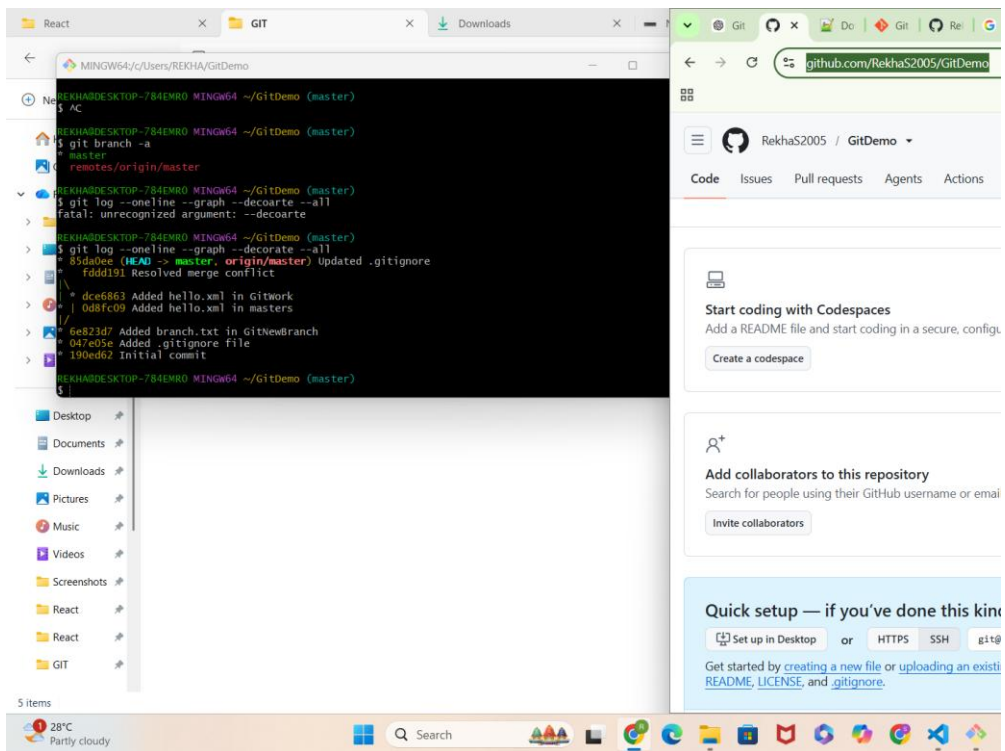
REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ ^C

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ ^C

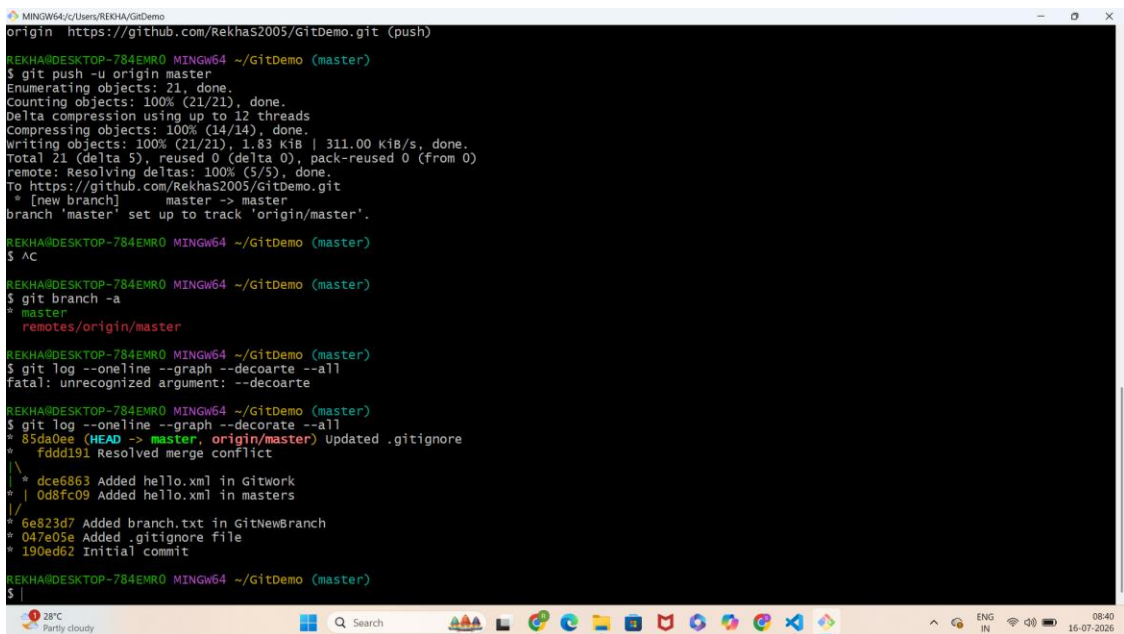
REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git remove origin
git: 'remove' is not a git command. See 'git --help'.
```

The terminal window has a taskbar at the bottom showing the Windows Start button, a search bar, and several application icons including File Explorer, Edge, and VS Code. The system tray shows the date and time as 28°C Partly cloudy.

Repository verification before remote configuration.



Successful push to GitHub and branch verification.



GitHub repository showing the uploaded project.

Result

The local Git repository was successfully connected to the GitHub remote repository. All commits were pushed successfully, the master branch now tracks origin/master, and the repository was verified both locally and on GitHub.